Every Round Is a Release: Communication-Budgeted Federated Learning Over Constrained Links

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***Abstract*—Cross-silo federated learning (FL) between hospitals is bound by the uplink. Each synchronisation round is a scheduled transfer over an institutional WAN, a cellular edge slice or a satellite link, and under client-level differential privacy (DP) each transfer is also a release that the privacy accountant must charge. This article treats the learning job as a network tenant with a single operator-facing quota ρ on the mean fraction of the shared model transmitted per round, and presents TierFed, a scheduler that measures the cross-site dispersion of each parameter tier and solves, exactly and in under a millisecond, the integer programme that minimises a local-SGD drift bound under the quota while a per-site classifier head stays private. On a patient-deduplicated readmission benchmark (20 sites, 10 seeds, 1813 bit-reproducible runs), TierFed at $\rho = 0.25$ attains 0.6269 AUROC against 0.6215 for full-rate FedPer with 70% less upload, and 1.6 MB per site with sparse ternary compression; the ordering holds on four real institutions and with a 1,740,290-parameter model whose uploads, replayed over paced TCP, validate the analytic link-cost model to within 2.1%. Three negative findings are reported with equal weight. A fixed uniform period matches or exceeds TierFed without DP and tier differentiation is rarely active. Under DP, a schedule derived from client updates is itself an uncharged release, which disqualifies every adaptive scheduler including TierFed; at matched $\varepsilon \leq 8$ on 20 and 100 sites every eligible method is near chance, the sparsest fixed period is best, and a privacy-aware planner does not improve on it. Every number is generated from released logs.**

***Index Terms*—Federated learning, communication efficiency, network resource allocation, 6G edge intelligence, differential privacy, link emulation, clinical prediction, non-IID data.**

I. Introduction

FEDERATED learning allows several institutions to train one predictive model without exchanging records [1], [2]. In the cross-silo setting that matters for healthcare the participants are a handful of sites, each behind its own firewall and connected to the aggregation server by a link that the institution's network team controls [3], [4]. The learning job is one tenant among many on an uplink with a fixed capacity, a scheduled transfer window and an audit trail, and whether the link is an institutional WAN, a 5G/6G edge slice or a satellite backhaul, the operator's question is the same: how many bytes the job pushes per window, and whether it fits [5], [6].

The scheduling problem addressed here is the following. A server coordinates N sites over R rounds. In each round it may request some, all or none of the shared parameters; each site that is asked trains locally and returns its slice. The server must keep the mean fraction of the shared model transmitted per round within a quota ρ, keep the personalised accuracy of every site non-inferior to full-rate personalisation, and, where client-level differential privacy (DP) is required, remain within a budget (ε, δ). Four properties of the problem shape the design and its evaluation. Parameters do not drift apart at one rate: the layers nearest the input are shared across sites and change slowly, the classifier head is site-specific, and one synchronisation rate suits neither. The quota is a network quantity, which has to map onto a slice allocation or a transfer window through bits, rate and latency. Under client-level DP every transmitting round is a release into which noise is injected and for which the accountant charges, so the set of transmitting rounds is a privacy decision as much as a network one, and a schedule that depends on the private updates is itself a release. Finally, the value of a scheduler can only be established against strong round-sparse and full-rate baselines on identical seeds, with the same link model, compressor and accountant applied to all.

TABLE I
Positioning by method class (–: not addressed)

Class	Timing	Budget	Head	DP-eligible
Full-rate averaging [1], [7], [8]	every round	–	shared	yes
Partial personalisation [9]–[11]	every round	–	private	yes
Adaptive period [12]	one, from loss	–	shared	no
Lazy upload [13]	per site, from update	–	shared	no
Layer-wise adaptive [14], [15]	per layer, from discrepancy	–	shared	no
Fixed multi-rate period	per tier, fixed	implicit	private	yes
TierFed (this article)	per tier, from dispersion	ρ, exact	private	no
TierFed-DP (this article)	per tier, planned	ρ and ε	private	yes

Prior work addresses these properties separately. Compression [16], [17] shrinks each upload but sends one every round. Local SGD with a fixed period [18], [19], AdaComm's loss-driven period [12] and LAG's lazy uploads [13] send less often, with one period for the whole model and no explicit budget. Partial personalisation keeps a head or a representation local [9]–[11] and synchronises the rest every round. Layer-wise schedules assign per-layer intervals from measured discrepancy [14] or learn per-layer weights [15]. FL over wireless links has been treated through joint learning-and-communication design [6], over-the-air computation [20] and 6G roadmaps [21]. Client-level DP follows McMahan *et al.* [22] with Rényi accounting [23], [24]; inversion [25], [26] and membership inference [27] measure what an adversary recovers. Table I summarises the gap: no published scheduler

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Code, per-seed logs and the scripts that generate every table, figure and number of this article are available at https://github.com/Abraheem13/tierfed.

accepts a byte quota, chooses per-tier intervals from measurements, withholds the head, and is evaluated under one link model and one DP accountant.

The contributions of this article are as follows.

1) **A budget-optimal multi-rate scheduler** (Section III). TierFed partitions the model into tiers, measures a scale-free dispersion proxy per tier from the slices it receives, and solves exactly, in under a millisecond, the integer programme that minimises the drift term $\sum_g K_g^2 \hat{\Gamma}_g$ of a blockwise local-SGD bound subject to $\sum_g w_g/K_g \le \rho$, with the classifier head never transmitted; ρ is the only operator-facing quantity and maps onto bytes, time and energy through a stated link model.
2) **Schedules as privacy releases** (Sections III-D and V-E). The standard client-level Gaussian accountant is shown not to cover a schedule computed from client updates, which disqualifies adaptive TierFed, AdaComm, LAG and FedLAMA under DP. A data-independent planner, TierFed-DP, chooses the period and the noise multiplier before training from (ε, δ), the sampling rate and ρ; the eligible schedules are compared at matched (ε, δ) with Poisson sampling on 20 and 100 sites, and the result is negative for every method and for the planner.
3) **Network costing validated by transport** (Section V-D). Every method is costed on five link profiles from its measured bits, and the model is checked by replaying each run's per-site uploads over paced loopback TCP with a 1,740,290-parameter model.
4) **A reproducible, paired evaluation** (Sections II and IV). A patient-deduplicated readmission benchmark with 20 specialty sites, four real institutions, an imaging collection and federations of 10 to 100 sites; nine baselines on one architecture table; ten seeds; Holm-corrected paired tests, BCa intervals and non-inferiority on AUROC and AUPRC; bit-identical reruns; every number regenerated from released logs by one command.

The evaluation also records where the scheduler does *not* help: a fixed uniform period of four rounds is the most accurate configuration tested, AdaComm is cheaper at equal accuracy without DP, tier differentiation is rarely active, no method separates from the others under inversion or membership inference, and a negative claim about fixed schedules made in an earlier version of this work is retracted.

II. System Model and Study Design

This section fixes the setting in which the scheduler acts: the federation and its tiers, the link model, the data and the metrics by which every method is judged.

A. Federation, Tiers and Multi-Rate Aggregation

N sites hold private datasets $\mathcal{D}_k$ of size n_k, $n = \sum_k n_k$, $p_k = n_k/n$. The shared objective is $F(\theta) = \sum_k p_k F_k(\theta)$ with $F_k(\theta) = \mathbb{E}_{\xi\sim\mathcal{D}_k}[f(\theta;\xi)]$. In round r each participating site initialises the shared part of its model from the broadcast state, performs one local epoch of $\tau_k = \lceil n_k/B \rceil$ optimiser steps with batch size B, and returns the shared slice that is due. Parameters are partitioned into disjoint tiers $\mathcal{G} = \{\mathrm{s}, \mathrm{m}, \mathrm{f}\}$ (slow, medium, fast) by their position in the forward computation graph (input projection, hidden blocks, head), with the classifier head always in the fast tier. The fast tier is private, so FedPer-style personalisation is the special case $K_\mathrm{s} = K_\mathrm{m} = 1$. With $\mathcal{A}_r \subseteq \mathcal{G}$ the tiers due in round r and $\mathcal{S}_r$ the participating sites, the server aggregates the active tiers only,

$$\theta_g^{(r+1)} = \begin{cases} \sum_{k\in\mathcal{S}_r} n_k\, \theta_{k,g}^{(r)} / \sum_{k\in\mathcal{S}_r} n_k, & g \in \mathcal{A}_r, \\ \theta_g^{(r)}, & g \notin \mathcal{A}_r, \end{cases} \tag{1}$$

and sites keep their own fast tier throughout. With w_g the share of the shared payload carried by tier g, the mean payload fraction over R rounds is

$$\Pi(\{K_g\}) = \frac{1}{R}\sum_{r=1}^{R} \sum_{g\in\mathcal{A}_r} w_g \;\longrightarrow\; \sum_{g\in\mathcal{G}_\mathrm{sh}} \frac{w_g}{K_g}, \tag{2}$$

where $\mathcal{G}_\mathrm{sh} = \{\mathrm{s}, \mathrm{m}\}$. Rounds with $\mathcal{A}_r = \emptyset$ transmit nothing; these local-only rounds are how a quota below $\min_g w_g$ is realised.

B. Links and Network Cost

The job is costed on a link profile (β, ℓ, P): uplink rate, round-trip latency and transmit power. Sites upload in parallel on independent links, so a round in which site k sends b_k bits costs

$$t_r = \max_k \frac{b_k}{\beta} + \ell, \qquad E_r = \sum_k \frac{b_k}{\beta} P, \tag{3}$$

and nothing when no site sends. Communication time and energy are summed over rounds; window compliance is the fraction of rounds whose t_r fits a scheduled window W. Five profiles are used (Table II): cell-edge 5G NR (5 Mbit/s, 30 ms, 0.2 W) [28], mid-cell NR (50 Mbit/s, 20 ms, 0.2 W) [28], a 6G edge-slice target (1 Gbit/s, 5 ms, 0.5 W) [21], an LEO satellite uplink (10 Mbit/s, 40 ms, 2 W) and a wired institutional WAN (100 Mbit/s, 10 ms), each with a 60-s window; they are representative targets, not measurements, and are the only assumptions in the network-level results. The model omits scheduling, fading and retransmission; it is a cost translation of measured bits, and Section V-D checks it against real transport.

C. Data, Sites and Metrics

Diabetes 130-US Hospitals [29] is the main benchmark: 30-day readmission from encounter-level features. Cleaning follows Strack *et al.* (encounters discharged to death or hospice removed, first encounter per patient kept), which leaves 55,992 training and 13,998 test encounters from 55,992 patients, 160 encoded features and a positive rate of 9.0%. The public release carries no hospital identifier, so sites are formed from the medical-specialty column by longest-processing-time bucketing with a client-share cap of 15% (a specialty above the cap is split, patient-grouped, before bucketing); the largest site holds 12% of the training data and the mean label total-variation distance between sites is 0.009. These clients are called *sites* rather than hospitals, and every split is patient-grouped. **Fed-Heart-Disease** [30], [31] pools the four UCI Heart Disease collections (Cleveland Clinic, Hungarian Institute of Cardiology, University Hospitals Zürich and Basel,

TABLE II
STUDY SETUP, READ FROM THE CONFIGURATION FILE AND THE RUN LOGS

Item	Value
Main data	Diabetes 130-US: 55,992/13,998 train/test encounters, 55,992 patients, 160 features, positive rate 9.0%, 20 specialty sites (share cap 15%)
Other data	Fed-Heart-Disease: 4 institutions, 736 training patients; PneumoniaMNIST: 5,232/624, 20 sites, Dirichlet $\alpha = 0.3$
Backbones	MLP 128–128–64 (45,506 parameters; tier shares 45.3/54.4/0.3%); tabular ResNet (width 128, depth 6); FT-Transformer (width 64, depth 3, 8 heads); wide MLP 1024–1024–512 (1,740,290); small CNN (imaging)
Training	60 rounds, one local epoch, batch 64, Adam 10^{-3} with persistent state, inverse-frequency class weighting, full participation unless stated
TierFed	$T_w = 5$, $K_{\max} = 8$, $\rho = 0.25$ unless stated; exact solver
Link profiles	NR cell edge 5 Mbit/s, 30 ms, 0.2 W; NR mid-cell 50, 20, 0.2; 6G slice 1000, 5, 0.5; LEO 10, 40, 2.0; WAN 100, 10, 0; window 60 s
Baselines	FedProx $\mu = 0.01$; FedLAMA-style $\tau = 5$, $\phi = 2$; AdaComm $\tau_0 = 8$; LAG $\xi = 1$; fixed period $(K_{\mathrm{s}}, K_{\mathrm{m}}) = (5, 2)$; published defaults, none tuned on test data
DP	$C = 1$, $\delta = 10^{-5}$, $\varepsilon^{\star} \in \{1, 2, 4, 8\}$, $q \in \{0.2, 0.5, 1\}$, σ calibrated per method; planner $\lambda = 1$ ($\{0.3, 3, 10\}$ as sensitivity)
Seeds, tests	10 seeds (5 for ResNet and Transformer, 3 for the wide model and scaling); paired, Holm-corrected; bit-identical reruns

VA Long Beach), one client per real institution, 736 training patients, target "any angiographic disease"; missingness indicators are added because two features are mostly missing outside Cleveland. The open eICU-CRD demo [32], [33] was prepared as well, but its largest hospital has 40 stays and the next 28, so it cannot support a per-hospital federation; the full eICU-CRD requires credentialed access and was not run. **PneumoniaMNIST** [34] is the imaging benchmark: a natively binary 28×28 collection partitioned by Dirichlet label skew ($\alpha = 0.3$) with a minimum site size of 50 images. Table II summarises the setup; every dataset statistic is read from the run logs.

Metrics. Because personalised methods hold private heads, every method is evaluated on each site's patient-grouped held-out split with that site's own parameters, sample-weighted (single-class splits are excluded identically for every method); reported are final-round federated AUROC and AUPRC, early-stopped and pooled AUROC, the global model's test AUROC, cumulative upload per site, transmitting rounds, and the costs of (3); Fig. 1 summarises the study.

Problem statement. Given a quota $\rho \in (0, 1]$ on the mean shared-payload fraction, and optionally a budget (ε, δ), choose the intervals $\{K_g\}$ such that each site's personalised accuracy is non-inferior to full-rate personalisation, the payload satisfies (2), the attack surface does not grow, and, under DP, the schedule itself reveals nothing about the private data.

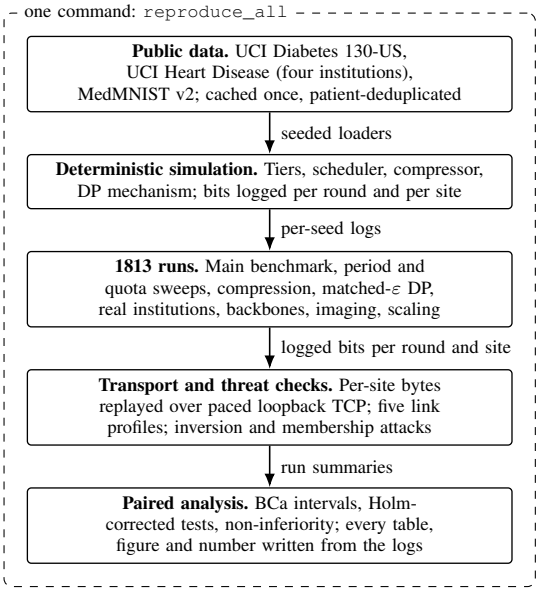


Fig. 1. Study pipeline from the public data to every reported number. All stages run from one command, and the determinism check runs first.

III. BUDGET-OPTIMAL MULTI-RATE SCHEDULING

The scheduler separates *what* is sent (tiers), *when* (per-tier intervals chosen under the quota), *how large* (an optional compressor) and *how private* (the DP mechanism and its accountant). Fig. 2 shows the architecture. The server-side controller observes the received slices, the sites observe only the broadcast, and the link and the privacy mechanism sit between them.

A. Dispersion Measurement

TierFed estimates the heterogeneity of each tier from the dispersion of the slices it actually receives. For tier g with tensors $j \in g$,

$$\hat{\Gamma}_g^{(r)} = \frac{\sum_{j \in g} \frac{1}{|\mathcal{S}_r|} \sum_{k \in \mathcal{S}_r} \left\| \theta_{k,j}^{(r)} - \bar{\theta}_j^{(r)} \right\|_2^2}{\sum_{j \in g} \left\| \bar{\theta}_j^{(r)} \right\|_2^2 + \epsilon}, \tag{4}$$

a scale-free ratio that makes tiers of different magnitude comparable. The estimate exists only for tiers received in round r (the last value is carried forward otherwise, with its age logged), and because the received slices result from $K_g \tau_k$ local steps it scales with the interval that produced it: $\hat{\Gamma}_g$ is a proxy for the drift the bound penalises, not an unbiased estimate of Γ_g.

B. The Integer Programme and Its Motivation

Bound. The analysis concerns the shared backbone with the private heads frozen; the personalisation gap this introduces is measured empirically (Section V-G). Under the standard local-SGD assumptions stated per tier, namely L-smooth F_k, unbiased stochastic gradients with per-tier variance σ_g^2, bounded per-tier heterogeneity $\frac{1}{N} \sum_k \|\nabla_g F_k(\theta) - \nabla_g F(\theta)\|^2 \le \Gamma_g$ and $F \ge F^{\star}$ [18], [19], the following holds.

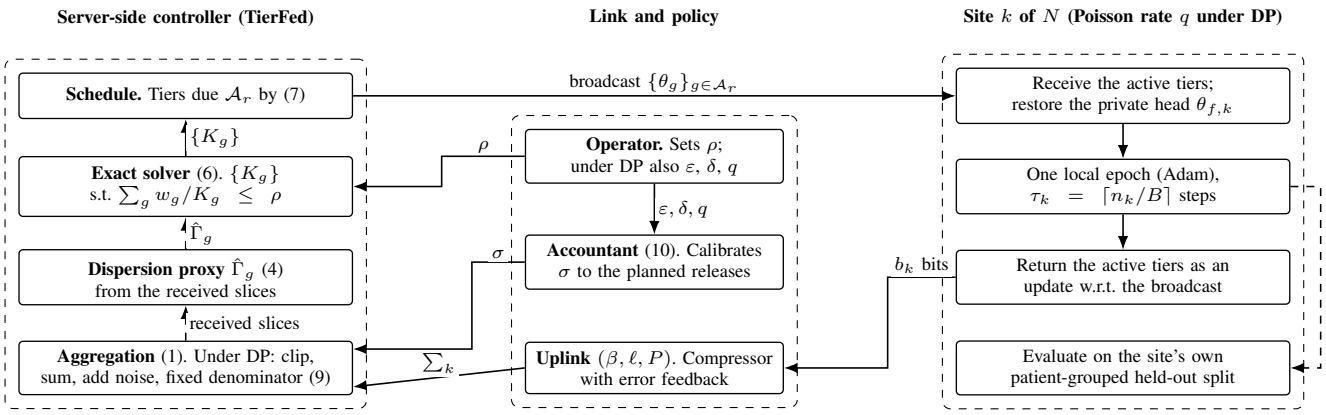


Fig. 2. Scheduler and environment. Solid arrows carry data and policy inputs; dashed arrows are measurements fed back. Under DP the dashed path from the received slices to the solver is cut and the schedule is planned before training (Section III-D).

Proposition 1 (Multi-rate local-SGD bound)**.** *With local SGD of step size η, $\tau_k \in [\tau_{\min}, \tau_{\max}]$ local steps per round and tier g averaged every K_g rounds, there are absolute constants c_0, c_1, c_2 such that for $\eta \le c_0/(LK_{\max}\tau_{\max})$,*

$$\frac{1}{R}\sum_{r=1}^{R}\mathbb{E}\big\|\nabla F(\bar{\theta}_r)\big\|^2 \le \frac{c_1(F(\bar{\theta}_1)-F^\star)}{\eta\tau_{\min}R} + c_1\eta L\sum_g \frac{\sigma_g^2}{N} + c_2\,\eta^2L^2\sum_g K_g^2\tau_{\max}^2\big(\Gamma_g+\sigma_g^2\big). \tag{5}$$

The proof applies the perturbed-iterate argument of [19] blockwise and is given in the supplement (Section S2). The cost of a long interval is quadratic in the interval and linear in heterogeneity, so a tier tolerates sparsity if and only if its divergence is small. The bound is for SGD; the benchmark uses Adam, the solver the baselines were introduced with, and an SGD arm (Section V-B) shows that the ordering of methods depends on the local solver. Nothing above proves that the programme below is the right one, only that its objective has the shape of an upper bound on the drift term.

Objective and constraint. Given a quota ρ the controller solves

$$\{K_g^\star\} = \arg\min_{K_g\in\{1,\dots,K_{\max}\}} \sum_{g\in\mathcal{G}_{\rm sh}} K_g^2\,\hat{\Gamma}_g \quad \text{s.t.} \sum_{g\in\mathcal{G}_{\rm sh}}\frac{w_g}{K_g} \le \rho. \tag{6}$$

Solver. With at most three shared tiers and $K_{\max} = 8$ the feasible set has at most $8^3 = 512$ points, so (6) is solved exactly by enumeration in well under a millisecond at every federation size tested (Section V-G). The greedy doubling rule of an earlier version of this work, which lengthens the interval of the tier with the smallest marginal bound cost per unit of bandwidth saved, $g^\star = \arg\min_g 6\,\hat{\Gamma}_g K_g^3/w_g$, reaches only powers of two and is reported as a baseline for the solver.

Scheduling rule and warm-up. Tier g is due when $r - \tau_g \ge K_g^\star$, with τ_g the round of its last synchronisation:

$$\mathcal{A}_r = \begin{cases} \mathcal{G}_{\rm sh}, & r \le T_w, \\ \{g:\ r-\tau_g \ge K_g^\star\}, & r > T_w. \end{cases} \tag{7}$$

Divergence is largest in the earliest rounds, so the first T_w rounds synchronise every shared tier. $T_w = 5$ and $K_{\max} = 8$ are held fixed across every dataset, architecture, loss and privacy setting; a T_w sweep is in Section V-G. Fig. 3 shows one server round.

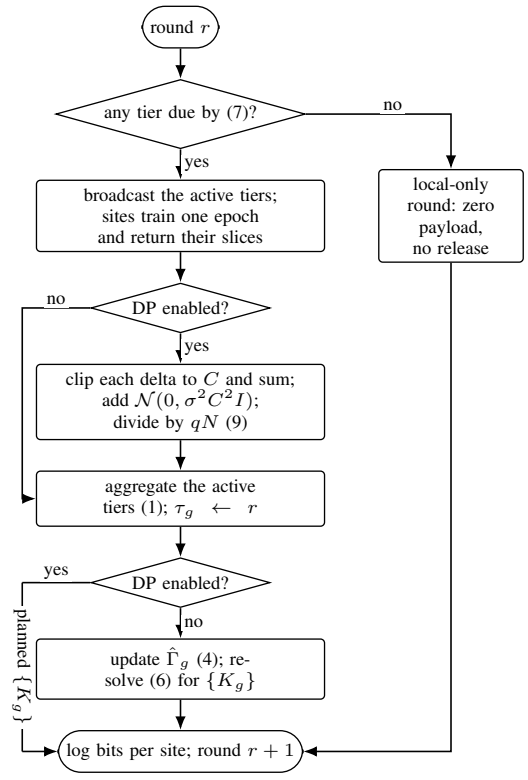


Fig. 3. One server round. Under DP the schedule is fixed before training and the dispersion feedback is never computed from the private slices.

C. Composition With Compression

TierFed controls when a tier is sent; a compressor controls how large the message is. For a compressor with rate $c \in (0, 1]$ bits per parameter relative to float32, the expected upload over R rounds is

$$B = R\,|\mathcal{S}|\,\Pi(\{K_g\})\,c\,32\,d_{\rm sh} \text{ bits}, \tag{8}$$

with d_{sh} the shared parameter count; the reduction factors multiply. Compressors act on each site's update relative to the broadcast with per-site error feedback, and communication is counted from what the compressor emitted for the sites that actually sent. The only operator-facing quantity is ρ. Through (8) and (3) it maps onto a slice allocation or a transfer-window budget, and Section V-D reports the mapping $\rho \mapsto$ (upload, time, energy) on every profile.

D. Privacy: Mechanism, Accountant and Eligible Schedules

Threat model. The adversary is an honest-but-curious aggregator or a passive eavesdropper on the uplink; both observe exactly the bytes a site transmits in each round, and therefore also *which* rounds it transmits in. Neither observes the private head, the local data or the optimiser state; poisoning is out of scope. The attack surface of a method is the sequence of slices it transmits, measured by update inversion [26] on the transmitted slices and by loss-threshold membership inference [27] on the deployed models.

Mechanism. Each round every site joins the cohort $\mathcal{S}_r$ independently with probability q (Poisson sampling; $q = 1$ is full participation). The server clips every participating site's returned delta and adds noise to the sum before dividing by the *fixed* expected cohort size,

$$\tilde{\Delta} = \frac{1}{qN}\Big(\sum_{k \in \mathcal{S}_r} \Delta_k \min\big(1, \tfrac{C}{\|\Delta_k\|_2}\big) + \mathcal{N}(0, \sigma^2 C^2 I)\Big), \quad (9)$$

the fixed-denominator DP-FedAvg estimator [22], so that adding or removing one site changes the sum by at most C in ℓ_2 and the denominator reveals nothing about the realised cohort; a planned release whose cohort happens to be empty still adds noise. The mechanism is applied by the server to every method identically.

Accountant. Over the R_{tx} planned releases the Rényi DP of the Poisson-subsampled Gaussian mechanism is composed at integer orders [24],

$$\varepsilon_\alpha = \frac{R_{\mathrm{tx}}}{\alpha - 1} \log \sum_{k=0}^{\alpha} \binom{\alpha}{k} (1-q)^{\alpha-k} q^k e^{\frac{k^2 - k}{2\sigma^2}},$$
$$\varepsilon = \min_\alpha \Big\{ \varepsilon_\alpha + \frac{\log(1/\delta)}{\alpha - 1} \Big\}, \quad (10)$$

with $\delta = 10^{-5}$; for $q = 1$ this reduces to $R_{\mathrm{tx}}\alpha/(2\sigma^2)$ [23]. The implementation reproduces the TensorFlow-Privacy MNIST reference ($\varepsilon \approx 3.0$) and the $q = 1$ closed form as unit tests. For a matched-privacy comparison every method receives the same target $\varepsilon^\star$ and σ is calibrated to its own release count by bisection on (10).

Which schedules are covered. The accountant covers a release only if the decision to release is independent of the private data. This holds for FedAvg, FedProx, FedPer, a fixed period and the planner below, whose release rounds are fixed before training. It does not hold for adaptive TierFed, AdaComm, LAG or FedLAMA: their schedules are functions of client updates or losses, so the release pattern carries information that (10) does not charge for, and the released code refuses to run them under DP. An earlier version of this article compared all methods at equal σ including the adaptive ones; that comparison is not covered by its stated budgets and is kept in the supplement with this caveat.

Planning under a privacy budget (TierFed-DP). A planner that respects the accountant must choose the schedule before training from public quantities only: the target $(\varepsilon^\star, \delta)$, the sampling rate q, the horizon R, the warm-up T_w, the tier shares w_g and the quota ρ. For every candidate $K \in \{1, \dots, K_{\max}\}^{|\mathcal{G}_{\mathrm{sh}}|}$ it computes the release set $\mathcal{R}(K) = \{r \le R : \ \exists g,\ r \le T_w \text{ or } (r - T_w) \bmod K_g = 0\}$, the smallest $\sigma(K)$ for which $|\mathcal{R}(K)|$ releases are $(\varepsilon^\star, \delta)$-DP, and the surrogate

$$J(K) = \frac{\sum_g w_g K_g^2}{K_{\max}^2} + \lambda \frac{E(K)}{E(\mathbf{1})},$$
$$E(K) = \frac{1}{R} \sum_{r \in \mathcal{R}(K)} \Big(\sum_{g \text{ due at } r} w_g \Big) \sigma(K)^2, \quad (11)$$

the drift term of (5) with a flat, data-free heterogeneity prior normalised by its worst case, plus the noise energy injected per round normalised by its full-rate value, so that $\lambda = 1$ weighs a fraction of worst-case drift against the same fraction of full-rate noise. Fewer releases need less noise per release but drift more; (11) trades the two subject to the byte quota, the minimiser is found by enumeration, and the server is told $\sigma(K)$. λ absorbs the unknown constants of the bound. Because the prior is flat the planner chooses equal intervals for both shared tiers; Section V-E evaluates it as an ablation and finds that it does not improve on the sparsest fixed period.

IV. Experimental Setup

A fair test needs strong alternatives, several regimes and many paired seeds. Ten experiment families are run, every one with every listed method on identical seeds: the main comparison (nine methods, 20 sites, 10 seeds); the period grid and the quota sweep $\rho \in [0.15, 0.8]$ with both solvers; baseline sweeps over AdaComm τ_0, LAG ξ and the SGD local solver; five compressors with FedAvg, FedPer and TierFed; matched-ε DP on 20 and 100 sites; inversion and membership attacks; the four heart-disease institutions; the wide model with three emulated profiles (3 seeds); the head $\times$ schedule, warm-up and loss ablations; and ResNet and FT-Transformer backbones (5 seeds), PneumoniaMNIST, and scaling from 10 to 100 sites (3 seeds). Table II lists the hyperparameters.

Baselines. *Full rate:* FedAvg [1], FedProx [7] ($\mu = 0.01$), SCAFFOLD [8] (control variates, twice the payload) and FedPer [9], the reference for every non-inferiority test. *Round-sparse:* AdaComm [12] ($\tau_0 = 8$; $\{2, 4, 16\}$ swept), LAG [13] ($\xi = 1$; up to 10^4 swept), a FedLAMA-style per-layer interval rule reimplemented from its published description [14] ($\tau = 5$, $\phi = 2$), and a fixed multi-rate period $(K_{\mathrm{s}}, K_{\mathrm{m}}) = (5, 2)$ with the same private head and warm-up as TierFed, which isolates the value of the measurement-driven choice of intervals. All methods share one architecture table, one evaluation protocol and one bit-accounting rule, a round in which no tier is due is local-only for the fixed period as for TierFed, and nothing was tuned on the test data.

Differential privacy. The five DP-eligible configurations, FedAvg, FedPer, fixed periods $K = 4$ and $K = 8$ and the planner with $\lambda = 1$, run at $C = 1$, $\delta = 10^{-5}$ on the 20

specialty sites at $\varepsilon^\star \in \{1, 2, 4, 8\}$, $q = 0.5$ and at $\varepsilon^\star = 4$, $q = 1$, and on a 100-site Dirichlet ($\alpha = 1$) federation of the same data at $\varepsilon^\star \in \{1, 2, 4, 8\}$, $q = 0.2$ and at $\varepsilon^\star = 4$, $q = 0.5$, so that cohorts of 10, 20 and 50 sites are covered. Each method's σ is calibrated to its own planned release count, and final-round AUROC is reported.

Link emulation. The emulation checks (3) against real transport. Each run's logged per-site upload sizes are replayed, round by round, as concurrent loopback TCP connections, one per transmitting site, paced by a token bucket at the profile's uplink rate after a one-way delay of half the RTT; the round ends when the last byte arrives. This application-layer emulation includes TCP segmentation, socket buffering and operating-system scheduling, but not radio scheduling, fading, loss or the downlink; time-to-target adds the measured compute time of the slowest site per round.

Determinism and statistics. Repeated runs of one configuration are bit-identical on the machine that produced the logs, and the reproduction script asserts this before any run. All comparisons are paired across the shared seeds. The reported statistics are means with BCa bootstrap 95% intervals, Holm-corrected paired t-tests over the family of all method-versus-FedPer contrasts with the paired standardised difference, and a one-sided non-inferiority test at margin $\delta_{\mathrm{NI}} = 0.01$ on both final AUROC and final AUPRC,

$$t_{\mathrm{NI}} = \frac{\bar{d} + \delta_{\mathrm{NI}}}{s_d/\sqrt{n}}, \qquad p = 1 - \mathcal{T}_{n-1}(t_{\mathrm{NI}}). \tag{12}$$

Parity is never inferred from a non-significant difference, and the margin is a reporting convention (0.01 AUROC is below the between-seed standard deviation of every method in Table III) with no clinical derivation.

V. Results

Every table body and number is generated from the per-seed logs. Unless stated otherwise, values are means over ten seeds on Diabetes with the MLP backbone, the best value in each table or group is set in bold, p-values are from paired tests across seeds, and tables numbered S-n are in the supplement.

A. Main Comparison

Table III compares the nine methods. Full-rate FedPer reaches 0.6215 AUROC at 217.8 MB per site. TierFed at $\rho = 0.25$ reaches 0.6269 at 65.3 MB, transmitting in 18 of 60 rounds; the paired difference to FedPer is +0.0054 (Holm-corrected $p = 0.006$; all eight contrasts are in Table S-XIV), and TierFed is non-inferior on AUROC ($p < 10^{-4}$) and on AUPRC ($p < 10^{-4}$). The round-sparse baselines are at least as accurate: AdaComm reaches 0.6302 at 29.1 MB (TierFed minus AdaComm -0.0033, $p = 0.024$) and the fixed period $(5, 2)$ reaches 0.6339 at 82.9 MB. Every sparse schedule is above full-rate FedPer after Holm correction, FedAvg and LAG (which never skips an upload at its default threshold) are level with it, and only SCAFFOLD with the Adam local solver is below (0.5930; with its native SGD solver it recovers to 0.6383, Table S-II). Fig. 4(a) and (b) show the trajectories: the sparse schedules reach the accuracy of FedPer within the warm-up and hold it while uploading a fraction of the bytes. Panel (c) shows the exact solver settling at $K_{\mathrm{s}} = K_{\mathrm{m}} = 4$ after warm-up while the dispersion of both tiers decays.

B. Periods, Quota and Solver

The fixed schedule's period sweep (Table S-I) shows that a uniform period of four rounds for both shared tiers reaches 0.6407 at 58.1 MB, the highest accuracy of any configuration in this study; only a long slow-tier period costs accuracy (0.6130 at $(8, 1)$), and the sparsest uniform period $(8, 8)$ keeps 0.6250 at 29.0 MB. Table IV shows that ρ behaves as an operator knob. Upload rises monotonically with the quota from 45.7 MB at $\rho = 0.15$ to 167.2 MB at $\rho = 0.8$, accuracy is flat above $\rho = 0.25$ and drops only at $\rho = 0.15$ (0.6070), and the best quota is $\rho = 0.4$ (0.6366 at 84.9 MB). The last column is the direct test of the measurement-driven mechanism: at $\rho = 0.25$ the solver gave the two shared tiers different intervals in 0% of post-warm-up rounds, at $\rho = 0.4$ in 7%, and at most in 26% at $\rho = 0.15$. On this data the controller mostly reduces to a uniform period chosen to meet the quota, which is why a well-chosen fixed period performs as well. Where the solvers differ, exactness matters: at $\rho = 0.4$ the exact solver reaches 0.6366 at 84.9 MB against 0.6283 at 95.7 MB for greedy doubling (Table S-I), which reaches only power-of-two intervals. The baseline sweeps (Table S-II) give the alternatives their own frontier: AdaComm with $\tau_0 = 8$ is the most byte-efficient configuration without DP (0.6302 at 29.1 MB), LAG skips uploads only at $\xi = 10^4$, where its accuracy collapses, and with an SGD local solver FedPer improves (0.6379) while TierFed is lower (0.6251). Fig. 5(a) shows the sweep with its intervals against the full-rate and fixed-period references.

C. Scheduling and Compression Compose

Table V confirms (8): timing and message-size reductions multiply. 8-bit quantisation takes FedPer from 217.8 to 54.5 MB at unchanged accuracy and TierFed to 16.4 MB (0.6266); with sparse ternary compression TierFed uploads 1.6 MB at 0.6290, against 5.2 MB (0.6250) for FedPer. Count-sketch compression fails for every method (0.4986, 0.5017, 0.5049): at the sketch width used the unsketched update is too noisy for this model, and the result is reported rather than tuned away.

D. Network Cost and a Wide Model on Emulated Links

On the five link profiles (Table S-III) the total communication time of the main case on a cell-edge NR uplink falls from 19.2 s for FedPer to 5.8 s for TierFed, with energy in proportion; every method fits the 60-s window because the model is small. With 1,740,290 parameters a full-rate upload is several megabytes per site and the link dominates the round on every constrained profile (Table S-X). The analytic model agrees with real paced TCP to within 2.1% on every one of the 35 emulated runs, so the costs are not an artefact of the formula; the residual is the RTT term, which the model charges in full and the emulator as a one-way delay, plus transport overhead. Over 60 rounds on cell-edge NR the emulated

TABLE III
RESULTS IN THE MAIN CASE (DIABETES 130-US, 20 SITES, 60 ROUNDS, 10 SEEDS). FINAL-ROUND FEDERATED AUROC WITH BCA 95% INTERVAL, AUPRC, EARLY-STOPPED AUROC, POOLED AUROC, CUMULATIVE UPLOAD PER SITE, TRANSMITTING ROUNDS, AND ONE-SIDED NON-INFERIORITY p AGAINST FEDPER AT $\delta_{\mathrm{NI}} = 0.01$ ON AUROC AND AUPRC

Method	AUROC [95% CI]	AUPRC	ES-AUROC	Pooled	Upload (MB)	Tx rounds	$\mathrm{NI}_{\mathrm{AUROC}}$ p	$\mathrm{NI}_{\mathrm{AUPRC}}$ p
FedAvg	0.6203 [0.6150, 0.6269]	0.171	0.6417	0.6209	218.4	60	$< 10^{-4}$	$< 10^{-4}$
FedProx	0.6279 [0.6229, 0.6359]	0.175	0.6433	0.6290	218.4	60	$< 10^{-4}$	$< 10^{-4}$
SCAFFOLD	0.5930 [0.5831, 0.5997]	0.138	0.6119	0.5930	436.9	60	0.997	1.000
FedPer	0.6215 [0.6170, 0.6274]	0.171	0.6419	0.6201	217.8	60	(ref.)	(ref.)
FedLAMA	0.6291 [0.6246, 0.6374]	0.176	0.6421	0.6295	154.1	60	$< 10^{-4}$	$< 10^{-4}$
AdaComm	0.6302 [0.6267, 0.6353]	0.168	0.6365	0.6290	29.1	8	$< 10^{-4}$	< 0.001
LAG	**0.6351** [0.6282, 0.6429]	0.179	0.6407	0.6362	218.4	60	$< 10^{-4}$	$< 10^{-4}$
Fixed schedule	0.6339 [0.6287, 0.6407]	0.177	0.6395	0.6338	82.9	37	$< 10^{-4}$	$< 10^{-4}$
TierFed	0.6269 [0.6234, 0.6342]	0.175	0.6415	0.6261	65.3	18	$< 10^{-4}$	$< 10^{-4}$

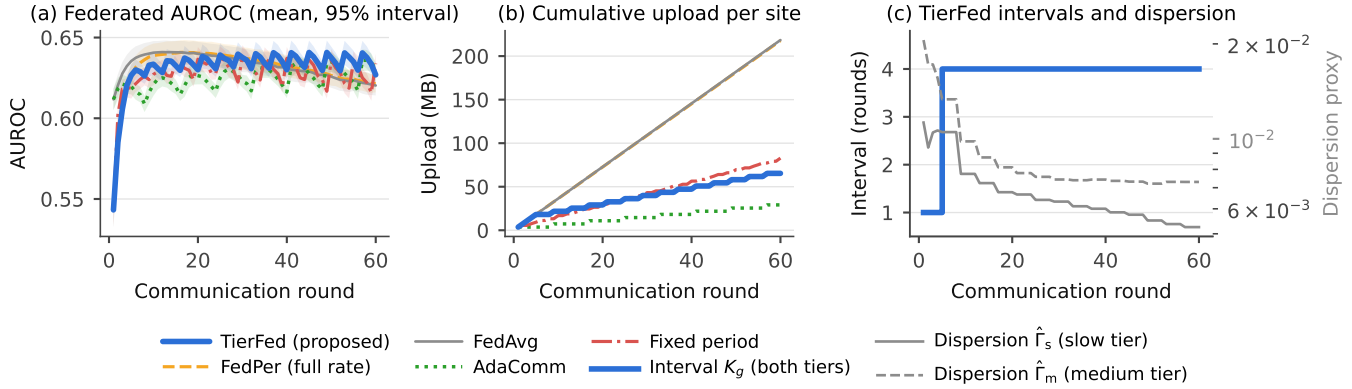


Fig. 4. Main case (10 seeds). (a) Federated AUROC per round with 95% intervals, (b) cumulative upload per site, and (c) the interval chosen by the exact solver at $\rho = 0.25$, identical for both shared tiers after warm-up, with the dispersion proxy (4) of each tier (log scale, right axis).

TABLE IV
QUOTA SWEEP FOR TIERFED (10 SEEDS): ACCURACY, UPLOAD, REALISED PAYLOAD FRACTION, MEAN INTERVALS CHOSEN BY THE EXACT SOLVER AFTER WARM-UP, AND SHARE OF POST-WARM-UP ROUNDS IN WHICH THE TWO SHARED TIERS RECEIVED DIFFERENT INTERVALS

ρ	AUROC	AUPRC	Upload (MB)	Realised	$\bar{K}_s$	$\bar{K}_m$	Diff. (%)
0.15	0.6070 ± 0.0119	0.163	45.7	0.21	6.8	6.4	26
0.25	0.6269 ± 0.0082	0.175	65.3	0.30	3.8	3.8	0
0.40	**0.6366** ± 0.0119	0.179	84.9	0.39	2.9	2.8	7
0.60	0.6290 ± 0.0093	0.176	116.2	0.53	1.9	1.9	0
0.80	0.6296 ± 0.0094	0.176	167.2	0.77	1.8	1.1	93

TABLE V
SCHEDULING × COMPRESSION (10 SEEDS): FINAL AUROC AND CUMULATIVE UPLOAD PER SITE (MB) FOR EACH COMPRESSOR, WHICH ACTS ON THE UPDATE WITH PER-SITE ERROR FEEDBACK; TIERFED AT $\rho = 0.25$

	FedAvg		FedPer		TierFed	
Compressor	AUROC	MB	AUROC	MB	AUROC	MB
None (float32)	0.6203	218.4	0.6215	217.8	**0.6269**	65.3
8-bit quantisation	0.6203	54.7	0.6213	54.5	**0.6266**	16.4
4-bit quantisation	0.6201	27.4	0.6223	27.3	**0.6265**	8.2
Top-k (1%)	0.6216	31.5	0.6227	31.4	**0.6277**	9.4
Sparse ternary (1%)	0.6234	5.2	0.6250	5.2	**0.6290**	1.6
Count sketch	0.4986	65.5	0.5017	65.3	**0.5049**	19.6

upload time is 669 s for FedPer, 201 s for TierFed, 178 s for the fixed period and 100 s for AdaComm. Time to a low target (AUROC 0.60) is not reduced by TierFed because the target is reached during the full-rate warm-up; the saving is in the remaining rounds (Fig. 5(d)), and AdaComm never reaches 0.62 within 60 rounds. On accuracy the wide model favours sparse synchronisation more strongly than the small one: the fixed period reaches 0.6352, AdaComm 0.6134 and TierFed 0.5999, against 0.5843 for FedPer and 0.5802 for FedAvg (3 seeds, indicative).

E. Privacy: Matched Budgets and Attack Surface

Matched budgets. Table VI, Tables S-XI and S-XII and Fig. 5(b),(c) compare the eligible methods at equal privacy rather than equal noise, and the result is negative for every method. With a cohort of 10 sites per round (20 sites, $q = 0.5$) the noise required at $\varepsilon^\star \le 8$ reduces every method to between 0.5073 and 0.5460 AUROC on a task whose non-private ceiling is 0.6215; with a cohort of 20 (100 sites, $q = 0.2$) the range is 0.5062 to 0.5217, and with a cohort of 50 the best method reaches 0.5421. The sensitivity of the sum is C regardless of the cohort, so the noise on the average scales as $1/(qN)$, and amplification by sampling lowers σ at equal ε without changing the conclusion: client-level DP on a federation of this size is noise-bound at any communication rate.

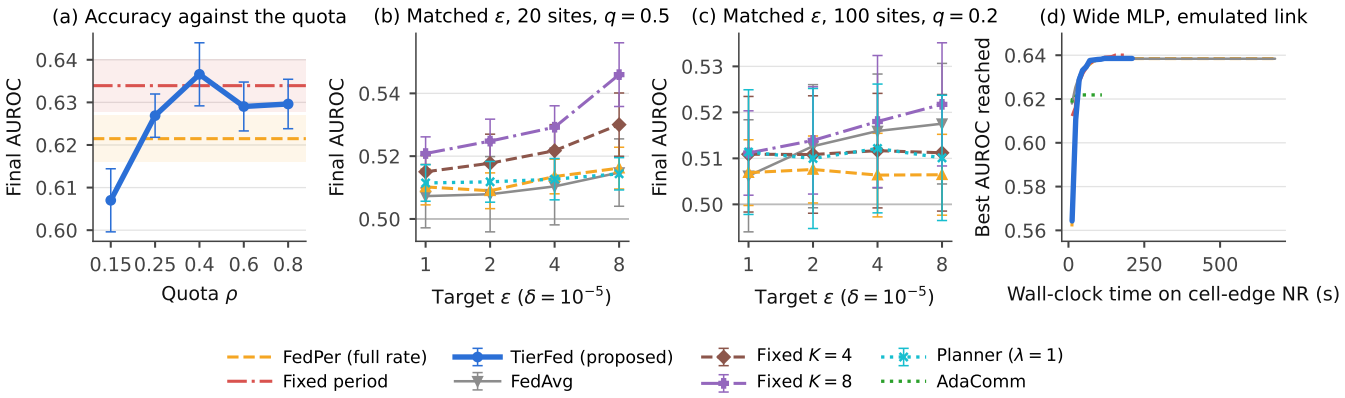


Fig. 5. (a) Final AUROC of TierFed against the quota ρ (mean and 95% interval over 10 seeds), with full-rate FedPer and the fixed period $(5, 2)$ as references. (b), (c) Final AUROC against the target ε at matched privacy on 20 and 100 sites: fewer releases help at every budget, and no method is far from chance at $\varepsilon \leq 8$. (d) Best federated AUROC reached by the wide model against wall-clock time on cell-edge NR, where each round costs the compute of the slowest site plus the upload time of (3).

TABLE VI
MATCHED-PRIVACY COMPARISON ON THE 20 SPECIALTY SITES (10 SEEDS, $C = 1$, $\delta = 10^{-5}$). EVERY METHOD RUNS AT THE SAME TARGET $\varepsilon^\star$ WITH POISSON SAMPLING AT RATE q; ITS σ IS CALIBRATED TO ITS OWN NUMBER OF PLANNED RELEASES BY (10), SO THE ACHIEVED ε EQUALS $\varepsilon^\star$ IN EVERY ROW. FINAL-ROUND AUROC FOR THE $q = 0.5$ SWEEP; THE $q = 1$ ROWS AND THE 100-SITE FEDERATION ARE IN TABLES S-XI AND S-XII

q, $\varepsilon^\star$, method	σ	Releases	AUROC
0.5, 1, FedAvg	19.14	60	0.5073 ± 0.0163
0.5, 1, FedPer	19.14	60	0.5102 ± 0.0092
0.5, 1, Fixed $K = 4$	10.95	19	0.5150 ± 0.0083
0.5, 1, Fixed $K = 8$	8.83	12	**0.5208** ± 0.0086
0.5, 1, TierFed-DP	12.00	23	0.5115 ± 0.0094
0.5, 2, FedAvg	9.84	60	0.5079 ± 0.0193
0.5, 2, FedPer	9.84	60	0.5090 ± 0.0092
0.5, 2, Fixed $K = 4$	5.72	19	0.5178 ± 0.0149
0.5, 2, Fixed $K = 8$	4.66	12	**0.5248** ± 0.0113
0.5, 2, TierFed-DP	6.25	23	0.5118 ± 0.0105
0.5, 4, FedAvg	5.18	60	0.5104 ± 0.0198
0.5, 4, FedPer	5.18	60	0.5135 ± 0.0089
0.5, 4, Fixed $K = 4$	3.09	19	0.5218 ± 0.0134
0.5, 4, Fixed $K = 8$	2.57	12	**0.5292** ± 0.0110
0.5, 4, TierFed-DP	3.36	23	0.5127 ± 0.0106
0.5, 8, FedAvg	2.84	60	0.5148 ± 0.0173
0.5, 8, FedPer	2.84	60	0.5162 ± 0.0107
0.5, 8, Fixed $K = 4$	1.76	19	0.5300 ± 0.0163
0.5, 8, Fixed $K = 8$	1.48	12	**0.5460** ± 0.0164
0.5, 8, TierFed-DP	1.89	23	0.5144 ± 0.0083

Within that regime the ordering is the one the release count predicts. The sparsest fixed period ($K = 8$, 12 releases) is the most accurate method in eight of the ten settings, and one of the two full-rate methods is the least accurate in every setting; at 20 sites and $\varepsilon^\star = 8$ the $K = 8$ period is +0.030 AUROC above FedPer ($p < 0.001$) and +0.016 above $K = 4$ ($p = 0.062$), and with a cohort of 50 it is +0.029 above FedPer ($p = 0.003$). The planner does not improve on this rule. With $\lambda = 1$ its surrogate (11) selects $K = 3$ (23 releases) on both federations and is level with FedPer and below $K = 8$ at every budget (20 sites, $\varepsilon^\star = 4$: -0.017, $p = 0.004$). The λ sweep (Table S-IV) shows that it responds to λ as designed, choosing K from 2 at $\lambda = 0.3$ to 5 at $\lambda = 10$, but no value improves on the rule: the best setting at 20 sites ($\lambda = 3$) reaches 0.5318 at $q = 1$, $\varepsilon^\star = 4$, within noise of the fixed periods there, was not chosen a priori, and gives the lowest planner AUROC of the sweep at 100 sites (0.5084). At these cohort sizes the noise term dominates so completely that the drift term only pulls the schedule towards more releases than are useful, so the planner is reported as an ablation, not as a method: under client-level DP on federations of this size the rule is to fix the sparsest schedule the accuracy target allows and to calibrate σ to its release count. The adaptive schedulers are absent by construction; the earlier equal-σ comparison that included them is in the supplement with that caveat.

Attack surface. Without DP neither attack separates the methods (Table S-V): inversion feature correlation is low for every method (TierFed 0.089, FedPer 0.088) and membership-inference AUC is close to chance (TierFed 0.540, FedPer 0.535). Per transmitted round TierFed exposes the same slice as FedPer; it only transmits in fewer rounds, which this generic attack does not exploit.

F. Real Institutions

On the four heart-disease institutions (Table S-XIII) TierFed reaches 0.8131 AUROC at 8.50 MB, significantly above FedPer (0.8016 at 28.32 MB; difference +0.012, $p = 0.009$) and level with FedAvg (+0.000, $p = 0.967$) and with the fixed period (0.8171 at 7.55 MB; $p = 0.221$); SCAFFOLD is the only method that is clearly worse. The ordering of the specialty-site benchmark therefore survives real institutional heterogeneity: sparse schedules match full-rate training at about a third of the upload, and the fixed period is as good as TierFed. The federation is small (736 training patients), so between-seed variance is large, and DP was not run on it.

G. Ablations, Backbones, Imaging and Scale

Ablations (Table S-VI). FedPer and FedAvg are level (0.6215 and 0.6203), so the private head does not matter on this data, and TierFed with and without the head is also

level (0.6269 and 0.6264). Warm-up length moves accuracy by at most 0.6354 at $T_w = 20$ against 0.6270 at $T_w = 0$, at a monotone cost in bytes, and TierFed matches or exceeds FedPer under every class-imbalance loss (LDAM: 0.6276 against 0.6150). An earlier version of this work reported that fixed multi-rate schedules lose about 0.05 AUROC to FedPer; that finding was produced by encounters of the same patient appearing in both training and test splits and by a fallback rule that sent the medium tier whenever no tier was due, both corrected here, and it does not reproduce.

Backbones (Table S-VII, 5 seeds). On every backbone the fixed period is at least as accurate as TierFed, and both at least as accurate as FedPer: on the ResNet TierFed reaches 0.6279 at 603.5 MB against 0.6137 at 2006.0 MB for FedPer, and on the FT-Transformer 0.6345 at 165.2 MB against 0.6352 at 581.2 MB.

Imaging (Table S-VIII). On PneumoniaMNIST the two methods that share the classifier head are the most accurate, FedAvg (0.9959) and the FedLAMA-style schedule (0.9965); TierFed reaches 0.9698 at 364.1 MB against 0.9643 at 1213.7 MB for FedPer ($p = 0.182$ for non-inferiority). Under strong label skew the private head is a liability, and a head-private schedule should not be used for such a task.

Scale (Table S-IX, 3 seeds). From 10 to 100 sites, upload per site scales linearly for every method, seconds per round are dominated by local training, the exact solver takes well under a millisecond at every size, and the ordering of methods lies within one standard deviation.

VI. Discussion

What the evidence supports. Three claims rest on ten-seed paired tests. Synchronising less does not cost accuracy on these benchmarks: every sparse schedule matches or exceeds full-rate personalisation, and TierFed is non-inferior to FedPer on AUROC and AUPRC at a fraction of the upload. Timing and message-size reductions compose multiplicatively with error-feedback compression, and the analytic translation of bytes into time and energy holds on real paced transport to within 2.1%. Under client-level DP the release schedule is part of the privacy mechanism: it must be fixed before training, and each release then carries noise set by the release count, so a communication quota and a privacy budget have to be planned together.

What it does not support. TierFed is not more accurate than simpler round-sparse alternatives: a fixed uniform period of four rounds is the most accurate configuration tested, AdaComm reaches comparable accuracy with fewer bytes when DP is off, and the measurement-driven mechanism rarely activates, since at most quotas the exact solver returns the same interval for both shared tiers. Under DP neither adaptive TierFed nor AdaComm is eligible, and the planned variant does not beat the sparsest fixed period at any value of its prior, because at cohorts of 10 to 50 sites the noise term leaves it nothing to trade; client-level DP on this task would need cohorts in the hundreds or a record-level mechanism, neither of which is tested here. The value of TierFed is therefore operational rather than statistical: it exposes a quota that maps directly onto a slice or window allocation, solves for the schedule exactly, and composes with compression and DP accounting, whereas AdaComm exposes a loss-driven period and a fixed schedule exposes nothing an operator can set from a byte budget. This is stated without qualification, and the earlier claim that fixed multi-rate schedules lose about 0.05 AUROC to personalised baselines is retracted (Section V-G).

Assumptions that bound the generality. *Sites.* The sites of the main benchmark are specialty buckets with a share cap, not hospitals; the real-institution federation has four members with a few hundred patients each. *Links.* The cost model and the socket emulation omit radio scheduling, fading, loss and the downlink, and the profiles are representative targets, not measurements. *Theory.* The bound is for SGD with full participation while the benchmark uses Adam, and $\hat{\Gamma}_g$ is a dispersion proxy whose feedback loop is observed to be stable but not proved so. *Privacy.* The inversion attack is generic, and client-level budgets of $\varepsilon \geq 1$ over 20 to 100 sites are weak guarantees. *Data.* The FT-Transformer arm used a non-deterministic attention kernel; under strong label skew on imaging the head-sharing methods win; and ten seeds on one tabular dataset, four institutions and one imaging collection are a benchmark, not a clinical validation.

Practical guidance. Set ρ from the transfer window through (8) and (3), compose the schedule with 8-bit quantisation or sparse ternary compression, and, where client-level DP is required, fix the schedule before training, calibrate σ to its release count, and avoid any scheduler that reacts to client updates.

VII. Conclusion

This article treated a cross-silo federated job as a network tenant with an explicit communication quota and measured, on bit-reproducible runs, how the number of transmitting rounds sets accuracy, network cost and privacy cost together. TierFed turns a byte quota into an exact per-tier schedule and is non-inferior to full-rate personalisation at a fraction of the upload, and the analytic link costs hold over paced TCP to within 2.1%. Two lessons follow. A well-chosen fixed period is as accurate as the measurement-driven schedule on this data, so the case for TierFed rests on the operator-facing quota and the exact solver, not on accuracy. Under client-level DP the schedule is itself a release: adaptive schedulers are not covered by the standard accountant, and at matched $\varepsilon \leq 8$ with cohorts of 10 to 50 sites every eligible method is near chance, the sparsest fixed period is best, and a privacy-aware planner does not improve on it. A communication quota and a privacy budget must therefore be planned jointly and before training; the evidence can be rerun from the released logs.

Data, Code and Declarations

Code, configuration, per-seed logs, emulation traces and the script that regenerates every table, figure and number are at https://github.com/Abraheem13/tierfed. All datasets are public and de-identified except the full eICU-CRD, which requires credentialed access; no ethics approval was required. Generative artificial-intelligence tools assisted with figure layout and

code refactoring; the author verified all content, code, numbers and figures. An earlier version, "Nested Federated Learning", is superseded by this article (supplement, Section S1). The author declares no competing interests and no funding.

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